

o Write a C program to simulate the following contiguous memory allocation techniques-

a) Worst-fit

b) Best-fit

c) First-fit

```
#include <stdio.h>
```

```
void allocation (int block[], int m, int process[], int n, int type
```

```
{
```

```
    int i, j, index;
```

```
    if (type == 1)
```

```
        printf ("In --- First Fit --- \n");
```

```
    else if (type == 2)
```

```
        printf ("In --- Best Fit --- \n");
```

```
    else
```

```
        printf ("In --- Worst Fit --- \n");
```

```
    printf ("Process No. \t Process Size \t Block No. \n");
```

```
    for (i = 0; i < n; i++)
```

```
    {
```

```
        index = -1;
```

```
        for (j = 0; j < m; j++)
```

```
            if (block[j] >= process[i])
```

```
            {
```

```
                if (type == 1)
```

```
                {
```

```
                    index = j;
```

```
                    break;
```

```
                }
```

```
            if (type == 2)
```

```
            {
```

```
                if (index == -1 || block[j] < block[index])
```

```
                    index = j;
```

```
            }
```


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```
printf("Enter sizes of %d processes: \n", n);
```

```
for (i = 0; i < n; i++)
```

```
{ scanf("%d", &p[i]); i++; }
```

```
allocation (b1, m, p, n, 1);
```

```
allocation (b2, m, p, n, 2);
```

```
allocation (b3, m, p, n, 3);
```

```
return 0;
```

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Output: Enter number of memory blocks: 5

Enter sizes of 5 memory blocks: 100 500 200 300 600

Enter number of processes: 4

Enter sizes of 4 processes:

212 417 112 426

--- First Fit ---

Process No.	Process Size	Block No.
1	212	2
2	417	5
3	112	2
4	426	Not Allocated

--- Best Fit ---

Process No. Process Size Block No.

1	212	4
2	417	2
3	112	3
4	426	5

--- Worst Fit ---

Process No. Process Size Block No.

1	212	5
2	417	2
3	112	5
4	426	5